```
import numpy as np
import os
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.ensemble import IsolationForest
from sklearn.metrics import confusion_matrix, classification_report
import time

from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report,
f1_score,roc_auc_score
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.callbacks import EarlyStopping
import random
from sklearn.model_selection import train_test_split
from tensorflow.keras import initializers
from sklearn.metrics import precision_score, recall_score, f1_score

seed = 42

os.environ['PYTHONHASHSEED'] = str(seed)
os.environ['TF_DETERMINISTIC_OPS'] = '1'

random.seed(seed)
np.random.seed(seed)
tf.random.set_seed(seed)
```

Build different Normalities

```
def load_timeseries_from_folders(normality, folder_list):
    all_dfs = []
    folders_to_read = folder_list[:normality]

    for folder in folders_to_read:
        # Full folder path
        folder_path = os.path.join("..", "TSB-UAD-Public", folder)

        # List all files in the folder
        files = [f for f in os.listdir(folder_path) if
os.path.isfile(os.path.join(folder_path, f))]

        if not files:
            print(f"Folder {folder} is empty or contains no files.")
            continue

        # Use the first file found
        filename = files[0]
```

```
        file_path = os.path.join(folder_path, filename)

        try:
            df = pd.read_csv(file_path, header=None).dropna()
            df.columns = ['measurement', 'anomaly']
            print(f"Read file {filename} from folder {folder}
({df.shape[0]} rows).")
            all_dfs.append(df)
        except Exception as e:
            print(f"Error reading file {file_path}: {e}")

    if all_dfs:
        full_df = pd.concat(all_dfs, ignore_index=True)
        print("\nData merge completed. Combine df
has",full_df.shape[0]," number of rows")
        return full_df
    else:
        print("No valid data found.")
        return pd.DataFrame()

# Example usage

folders = ["YAHOO","SensorScope","GHL","Genesis"]
normality = 1

result = load_timeseries_from_folders(normality, folders)
result
```

```
Read file YahooA3Benchmark-TS100_data.out from folder YAHOO (1680
rows).

Data merge completed. Combine df has 1680  number of rows
```

	measurement	anomaly
0	6.470061	0
1	80.972293	0
2	159.014680	0
3	214.721803	0
4	201.695834	0
...	...	...
1675	-143.374166	0
1676	-213.830741	0
1677	-191.906215	0
1678	-158.322475	0
1679	-89.296180	0

```
[1680 rows x 2 columns]
```

AE Offline

```
#parameters
window_size = 20
rolling_window = 10
max_epochs = 100

data_raw = result['measurement'].values
labels = result['anomaly'].values
n_samples = len(data_raw)

start_time = time.time()


#Features
rolling_mean = pd.Series(data_raw).rolling(window=rolling_window,
min_periods=1).mean().values
rolling_std = pd.Series(data_raw).rolling(window=rolling_window,
min_periods=1).std().fillna(0).values
features = np.column_stack([data_raw, rolling_mean, rolling_std])

#Scaling
scaler = StandardScaler()
features_scaled = scaler.fit_transform(features)

#Sliding Windows
def create_windows(data, window_size):
    return np.array([data[i:i+window_size] for i in range(len(data) -
window_size + 1)])

X_all = create_windows(features_scaled, window_size)
y_all = labels[window_size-1:]

#Train using normal only
X_train = X_all[y_all == 0]

X_train_train, X_val = train_test_split(X_train, test_size=0.1,
random_state=seed)


input_shape = (window_size, X_all.shape[2])
initializer = initializers.HeNormal(seed=seed)

model = models.Sequential([
    layers.Input(shape=input_shape),
    layers.Conv1D(32, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
    layers.Conv1D(16, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
```

```
    layers.Conv1DTranspose(16, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
    layers.Conv1DTranspose(32, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
    layers.Conv1D(X_all.shape[2], kernel_size=1, activation='linear',
padding='same', kernel_initializer=initializer)
])

model.compile(optimizer='adam', loss='mse')
early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True, verbose=0)

model.fit(
    X_train_train, X_train_train,
    validation_data=(X_val, X_val),
    epochs=max_epochs,
    batch_size=128,
    shuffle=False,
    callbacks=[early_stop],
    verbose=1
)


X_pred = model.predict(X_all, verbose=0)
mse = np.mean(np.square(X_all - X_pred), axis=(1, 2))


#Threshold selection
best_f1 = 0
best_thresh = np.percentile(mse, 90)

for p in range(80, 100):
    thresh = np.percentile(mse, p)
    y_pred = (mse > thresh).astype(int)
    f1 = f1_score(y_all, y_pred)
    if f1 > best_f1:
        best_f1 = f1
        best_thresh = thresh

final_preds = (mse > best_thresh).astype(int)

end_time = time.time()
elapsed_time = end_time - start_time


result['prediction'] = np.concatenate([np.zeros(window_size-1),
final_preds]).astype(int)
result['reconstruction_error'] = np.concatenate([np.zeros(window_size-
1), mse])
```

```
print("\n Results:")
print(classification_report(labels, result['prediction']))

print(f"Precision: {precision_score(labels,
result['prediction']):.4f}")
print(f"Recall:    {recall_score(labels, result['prediction']):.4f}")
print(f"F1 Score:  {f1_score(labels, result['prediction']):.4f}")
print(f"ROC-AUC:   {roc_auc_score(labels,
result['reconstruction_error']):.4f}")
print(f"runtime: {elapsed_time / 60:.4f} minutes")
```

```
Epoch 1/100
12/12 [==============================] - 2s 15ms/step - loss: 1.1346 -
val_loss: 0.4932
Epoch 2/100
12/12 [==============================] - 0s 4ms/step - loss: 0.3750 -
val_loss: 0.2505
Epoch 3/100
12/12 [==============================] - 0s 4ms/step - loss: 0.2155 -
val_loss: 0.1654
Epoch 4/100
12/12 [==============================] - 0s 4ms/step - loss: 0.1546 -
val_loss: 0.1275
Epoch 5/100
12/12 [==============================] - 0s 4ms/step - loss: 0.1223 -
val_loss: 0.1045
Epoch 6/100
12/12 [==============================] - 0s 4ms/step - loss: 0.1036 -
val_loss: 0.0900
Epoch 7/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0898 -
val_loss: 0.0796
Epoch 8/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0798 -
val_loss: 0.0714
Epoch 9/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0720 -
val_loss: 0.0648
Epoch 10/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0657 -
val_loss: 0.0593
Epoch 11/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0603 -
val_loss: 0.0543
Epoch 12/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0554 -
val_loss: 0.0498
Epoch 13/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0510 -
```

```
val_loss: 0.0458
Epoch 14/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0471 -
val_loss: 0.0424
Epoch 15/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0437 -
val_loss: 0.0394
Epoch 16/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0407 -
val_loss: 0.0366
Epoch 17/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0379 -
val_loss: 0.0342
Epoch 18/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0353 -
val_loss: 0.0319
Epoch 19/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0329 -
val_loss: 0.0298
Epoch 20/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0307 -
val_loss: 0.0278
Epoch 21/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0287 -
val_loss: 0.0260
Epoch 22/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0268 -
val_loss: 0.0245
Epoch 23/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0252 -
val_loss: 0.0231
Epoch 24/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0238 -
val_loss: 0.0219
Epoch 25/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0225 -
val_loss: 0.0208
Epoch 26/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0213 -
val_loss: 0.0197
Epoch 27/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0203 -
val_loss: 0.0188
Epoch 28/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0193 -
val_loss: 0.0179
Epoch 29/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0185 -
val_loss: 0.0172
```

```
Epoch 30/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0177 -
val_loss: 0.0165
Epoch 31/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0170 -
val_loss: 0.0158
Epoch 32/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0164 -
val_loss: 0.0152
Epoch 33/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0158 -
val_loss: 0.0147
Epoch 34/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0152 -
val_loss: 0.0142
Epoch 35/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0147 -
val_loss: 0.0137
Epoch 36/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0142 -
val_loss: 0.0132
Epoch 37/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0138 -
val_loss: 0.0128
Epoch 38/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0134 -
val_loss: 0.0124
Epoch 39/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0130 -
val_loss: 0.0120
Epoch 40/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0126 -
val_loss: 0.0117
Epoch 41/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0122 -
val_loss: 0.0114
Epoch 42/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0119 -
val_loss: 0.0111
Epoch 43/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0116 -
val_loss: 0.0108
Epoch 44/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0113 -
val_loss: 0.0105
Epoch 45/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0110 -
val_loss: 0.0102
Epoch 46/100
```

```
12/12 [==============================] - 0s 4ms/step - loss: 0.0107 -
val_loss: 0.0099
Epoch 47/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0105 -
val_loss: 0.0097
Epoch 48/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0102 -
val_loss: 0.0095
Epoch 49/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0100 -
val_loss: 0.0093
Epoch 50/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0098 -
val_loss: 0.0090
Epoch 51/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0096 -
val_loss: 0.0089
Epoch 52/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0093 -
val_loss: 0.0087
Epoch 53/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0092 -
val_loss: 0.0085
Epoch 54/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0090 -
val_loss: 0.0083
Epoch 55/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0088 -
val_loss: 0.0081
Epoch 56/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0086 -
val_loss: 0.0080
Epoch 57/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0085 -
val_loss: 0.0078
Epoch 58/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0083 -
val_loss: 0.0076
Epoch 59/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0081 -
val_loss: 0.0075
Epoch 60/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0080 -
val_loss: 0.0073
Epoch 61/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0078 -
val_loss: 0.0072
Epoch 62/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0077 -
```

```
val_loss: 0.0071
Epoch 63/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0076 -
val_loss: 0.0069
Epoch 64/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0074 -
val_loss: 0.0068
Epoch 65/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0073 -
val_loss: 0.0067
Epoch 66/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0072 -
val_loss: 0.0066
Epoch 67/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0071 -
val_loss: 0.0065
Epoch 68/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0069 -
val_loss: 0.0063
Epoch 69/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0068 -
val_loss: 0.0062
Epoch 70/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0067 -
val_loss: 0.0061
Epoch 71/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0066 -
val_loss: 0.0060
Epoch 72/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0065 -
val_loss: 0.0059
Epoch 73/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0064 -
val_loss: 0.0058
Epoch 74/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0063 -
val_loss: 0.0057
Epoch 75/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0062 -
val_loss: 0.0057
Epoch 76/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0061 -
val_loss: 0.0056
Epoch 77/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0060 -
val_loss: 0.0055
Epoch 78/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0059 -
val_loss: 0.0054
```

```
Epoch 79/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0058 -
val_loss: 0.0054
Epoch 80/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0057 -
val_loss: 0.0053
Epoch 81/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0056 -
val_loss: 0.0052
Epoch 82/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0056 -
val_loss: 0.0051
Epoch 83/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0055 -
val_loss: 0.0051
Epoch 84/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0054 -
val_loss: 0.0050
Epoch 85/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0053 -
val_loss: 0.0049
Epoch 86/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0052 -
val_loss: 0.0049
Epoch 87/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0052 -
val_loss: 0.0048
Epoch 88/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0051 -
val_loss: 0.0048
Epoch 89/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0050 -
val_loss: 0.0047
Epoch 90/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0050 -
val_loss: 0.0046
Epoch 91/100
12/12 [==============================] - 0s 5ms/step - loss: 0.0049 -
val_loss: 0.0046
Epoch 92/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0048 -
val_loss: 0.0045
Epoch 93/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0048 -
val_loss: 0.0045
Epoch 94/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0047 -
val_loss: 0.0044
Epoch 95/100
```

```
12/12 [==============================] - 0s 4ms/step - loss: 0.0047 -
val_loss: 0.0044
Epoch 96/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0046 -
val_loss: 0.0043
Epoch 97/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0045 -
val_loss: 0.0043
Epoch 98/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0045 -
val_loss: 0.0042
Epoch 99/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0044 -
val_loss: 0.0042
Epoch 100/100
12/12 [==============================] - 0s 4ms/step - loss: 0.0044 -
val_loss: 0.0041

 Results:
              precision    recall  f1-score   support

           0       1.00      0.99      1.00      1670
           1       0.35      0.60      0.44        10

    accuracy                           0.99      1680
   macro avg       0.68      0.80      0.72      1680
weighted avg       0.99      0.99      0.99      1680

Precision: 0.3529
Recall:    0.6000
F1 Score:  0.4444
ROC-AUC:   0.9937
runtime: 0.1289 minutes
```

AE NAIVE STREAMING

```
#parameters
window_size = 20
rolling_window = 10
batch_size = 500
max_epochs = 100

data_raw = result['measurement'].values
labels = result['anomaly'].values
n_samples = len(data_raw)

all_preds = np.zeros(n_samples)
all_mse = np.zeros(n_samples)
```

```
initializer = initializers.HeNormal(seed=seed)

def create_windows(data, window_size):
    X = []
    for i in range(len(data) - window_size + 1):
        X.append(data[i:i+window_size])
    return np.array(X)

num_batches = n_samples // batch_size + (1 if n_samples % batch_size !
= 0 else 0)

start_time = time.time()

for b in range(num_batches):
    print(f"\n Processing Batch {b+1}/{num_batches}")

    start_idx = b * batch_size
    end_idx = min(start_idx + batch_size, n_samples)
    batch_raw = data_raw[start_idx:end_idx]
    batch_labels = labels[start_idx:end_idx]

    #Rolling mean & std
    rolling_mean = pd.Series(batch_raw).rolling(window=rolling_window,
min_periods=1).mean().values
    rolling_std = pd.Series(batch_raw).rolling(window=rolling_window,
min_periods=1).std().fillna(0).values

    #Features
    batch_features = np.column_stack([batch_raw, rolling_mean,
rolling_std])

    #Scaling
    scaler = StandardScaler()
    batch_features_scaled = scaler.fit_transform(batch_features)

    #Sliding windows
    if len(batch_features_scaled) < window_size:
        print("Skip batch: not enough data for sliding windows")
        continue

    X_batch_all = create_windows(batch_features_scaled, window_size)
    y_batch_true = batch_labels[window_size-1:]

    #Train using normal only
    X_batch_train = X_batch_all[y_batch_true == 0]

    if len(X_batch_train) < 10:
        print("Skip batch: too few normal samples for training")
        continue
```

```
    input_shape = (window_size, X_batch_all.shape[2])

    model = models.Sequential([
        layers.Input(shape=input_shape),
        layers.Conv1D(32, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
        layers.Conv1D(16, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
        layers.Conv1DTranspose(16, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
        layers.Conv1DTranspose(32, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
        layers.Conv1D(X_batch_all.shape[2], kernel_size=1,
activation='linear', padding='same', kernel_initializer=initializer)
    ])

    model.compile(optimizer='adam', loss='mse')

    X_train_train, X_val = train_test_split(X_batch_train,
test_size=0.1, random_state=seed)
    early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True, verbose=0)

    model.fit(
        X_train_train, X_train_train,
        validation_data=(X_val, X_val),
        epochs=max_epochs,
        batch_size=128,
        shuffle=False,
        callbacks=[early_stop],
        verbose=1
    )


    X_pred = model.predict(X_batch_all, verbose=0)
    mse = np.mean(np.square(X_batch_all - X_pred), axis=(1, 2))

    #Threshold search
    best_f1 = 0
    best_thresh = np.percentile(mse, 90)

    for p in range(80, 100):
        threshold = np.percentile(mse, p)
        y_pred = (mse > threshold).astype(int)
        f1 = f1_score(y_batch_true, y_pred)
        if f1 > best_f1:
            best_f1 = f1
            best_thresh = threshold
```

```
    y_batch_pred = (mse > best_thresh).astype(int)


    batch_pred_full = np.zeros(end_idx - start_idx)
    batch_mse_full = np.zeros(end_idx - start_idx)

    batch_pred_full[window_size-1:] = y_batch_pred
    batch_mse_full[window_size-1:] = mse

    all_preds[start_idx:end_idx] = batch_pred_full
    all_mse[start_idx:end_idx] = batch_mse_full


end_time = time.time()
elapsed_time = end_time - start_time


result['prediction'] = all_preds.astype(int)
result['reconstruction_error'] = all_mse

print("\nResults:")
print(classification_report(labels, all_preds.astype(int)))

print(f"Precision: {precision_score(labels,
all_preds.astype(int)):.4f}")
print(f"Recall:    {recall_score(labels, all_preds.astype(int)):.4f}")
print(f"F1 Score:  {f1_score(labels, all_preds.astype(int)):.4f}")
print(f"ROC-AUC:   {roc_auc_score(labels, all_mse):.4f}")
print(f"runtime: {elapsed_time / 60:.4f} minutes")



 Processing Batch 1/4
Epoch 1/100
4/4 [==============================] - 0s 66ms/step - loss: 1.8839 -
val_loss: 1.1209
Epoch 2/100
4/4 [==============================] - 0s 8ms/step - loss: 1.0199 -
val_loss: 0.7014
Epoch 3/100
4/4 [==============================] - 0s 8ms/step - loss: 0.6617 -
val_loss: 0.5157
Epoch 4/100
4/4 [==============================] - 0s 8ms/step - loss: 0.4910 -
val_loss: 0.4061
Epoch 5/100
4/4 [==============================] - 0s 8ms/step - loss: 0.3879 -
val_loss: 0.3292
Epoch 6/100
4/4 [==============================] - 0s 8ms/step - loss: 0.3185 -
val_loss: 0.2717
```

```
Epoch 7/100
4/4 [==============================] - 0s 8ms/step - loss: 0.2676 -
val_loss: 0.2283
Epoch 8/100
4/4 [==============================] - 0s 8ms/step - loss: 0.2315 -
val_loss: 0.1993
Epoch 9/100
4/4 [==============================] - 0s 10ms/step - loss: 0.2075 -
val_loss: 0.1779
Epoch 10/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1884 -
val_loss: 0.1615
Epoch 11/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1732 -
val_loss: 0.1495
Epoch 12/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1611 -
val_loss: 0.1389
Epoch 13/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1493 -
val_loss: 0.1290
Epoch 14/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1386 -
val_loss: 0.1212
Epoch 15/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1302 -
val_loss: 0.1151
Epoch 16/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1231 -
val_loss: 0.1098
Epoch 17/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1167 -
val_loss: 0.1047
Epoch 18/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1107 -
val_loss: 0.0997
Epoch 19/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1052 -
val_loss: 0.0950
Epoch 20/100
4/4 [==============================] - 0s 9ms/step - loss: 0.1003 -
val_loss: 0.0908
Epoch 21/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0958 -
val_loss: 0.0872
Epoch 22/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0918 -
val_loss: 0.0840
Epoch 23/100
```

```
4/4 [==============================] - 0s 9ms/step - loss: 0.0881 -
val_loss: 0.0809
Epoch 24/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0848 -
val_loss: 0.0780
Epoch 25/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0817 -
val_loss: 0.0754
Epoch 26/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0788 -
val_loss: 0.0729
Epoch 27/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0762 -
val_loss: 0.0707
Epoch 28/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0737 -
val_loss: 0.0686
Epoch 29/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0714 -
val_loss: 0.0666
Epoch 30/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0691 -
val_loss: 0.0647
Epoch 31/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0670 -
val_loss: 0.0629
Epoch 32/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0651 -
val_loss: 0.0612
Epoch 33/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0632 -
val_loss: 0.0597
Epoch 34/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0615 -
val_loss: 0.0582
Epoch 35/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0598 -
val_loss: 0.0567
Epoch 36/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0582 -
val_loss: 0.0553
Epoch 37/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0566 -
val_loss: 0.0540
Epoch 38/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0551 -
val_loss: 0.0526
Epoch 39/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0537 -
```

```
val_loss: 0.0514
Epoch 40/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0523 -
val_loss: 0.0501
Epoch 41/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0509 -
val_loss: 0.0489
Epoch 42/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0496 -
val_loss: 0.0478
Epoch 43/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0483 -
val_loss: 0.0466
Epoch 44/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0471 -
val_loss: 0.0456
Epoch 45/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0458 -
val_loss: 0.0445
Epoch 46/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0447 -
val_loss: 0.0434
Epoch 47/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0435 -
val_loss: 0.0424
Epoch 48/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0424 -
val_loss: 0.0414
Epoch 49/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0413 -
val_loss: 0.0404
Epoch 50/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0402 -
val_loss: 0.0395
Epoch 51/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0392 -
val_loss: 0.0385
Epoch 52/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0382 -
val_loss: 0.0376
Epoch 53/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0372 -
val_loss: 0.0368
Epoch 54/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0363 -
val_loss: 0.0359
Epoch 55/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0354 -
val_loss: 0.0351
```

```
Epoch 56/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0345 -
val_loss: 0.0343
Epoch 57/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0337 -
val_loss: 0.0335
Epoch 58/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0329 -
val_loss: 0.0328
Epoch 59/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0321 -
val_loss: 0.0320
Epoch 60/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0313 -
val_loss: 0.0313
Epoch 61/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0306 -
val_loss: 0.0306
Epoch 62/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0298 -
val_loss: 0.0299
Epoch 63/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0291 -
val_loss: 0.0293
Epoch 64/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0285 -
val_loss: 0.0287
Epoch 65/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0278 -
val_loss: 0.0281
Epoch 66/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0272 -
val_loss: 0.0276
Epoch 67/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0266 -
val_loss: 0.0270
Epoch 68/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0261 -
val_loss: 0.0266
Epoch 69/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0255 -
val_loss: 0.0261
Epoch 70/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0250 -
val_loss: 0.0256
Epoch 71/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0245 -
val_loss: 0.0252
Epoch 72/100
```

```
4/4 [==============================] - 0s 9ms/step - loss: 0.0241 -
val_loss: 0.0248
Epoch 73/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0237 -
val_loss: 0.0245
Epoch 74/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0232 -
val_loss: 0.0241
Epoch 75/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0228 -
val_loss: 0.0237
Epoch 76/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0224 -
val_loss: 0.0233
Epoch 77/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0221 -
val_loss: 0.0230
Epoch 78/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0217 -
val_loss: 0.0227
Epoch 79/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0214 -
val_loss: 0.0223
Epoch 80/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0210 -
val_loss: 0.0220
Epoch 81/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0207 -
val_loss: 0.0217
Epoch 82/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0204 -
val_loss: 0.0214
Epoch 83/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0201 -
val_loss: 0.0211
Epoch 84/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0198 -
val_loss: 0.0208
Epoch 85/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0195 -
val_loss: 0.0206
Epoch 86/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0192 -
val_loss: 0.0203
Epoch 87/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0190 -
val_loss: 0.0200
Epoch 88/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0187 -
```

```
val_loss: 0.0198
Epoch 89/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0184 -
val_loss: 0.0195
Epoch 90/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0182 -
val_loss: 0.0193
Epoch 91/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0179 -
val_loss: 0.0190
Epoch 92/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0177 -
val_loss: 0.0188
Epoch 93/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0175 -
val_loss: 0.0186
Epoch 94/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0173 -
val_loss: 0.0184
Epoch 95/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0170 -
val_loss: 0.0181
Epoch 96/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0168 -
val_loss: 0.0179
Epoch 97/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0166 -
val_loss: 0.0177
Epoch 98/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0164 -
val_loss: 0.0175
Epoch 99/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0162 -
val_loss: 0.0173
Epoch 100/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0160 -
val_loss: 0.0171

 Processing Batch 2/4
Epoch 1/100
4/4 [==============================] - 0s 62ms/step - loss: 1.7766 -
val_loss: 1.1045
Epoch 2/100
4/4 [==============================] - 0s 10ms/step - loss: 0.9568 -
val_loss: 0.6884
Epoch 3/100
4/4 [==============================] - 0s 9ms/step - loss: 0.6170 -
val_loss: 0.4967
Epoch 4/100
```

```
4/4 [==============================] - 0s 9ms/step - loss: 0.4475 -
val_loss: 0.3831
Epoch 5/100
4/4 [==============================] - 0s 10ms/step - loss: 0.3460 -
val_loss: 0.3080
Epoch 6/100
4/4 [==============================] - 0s 10ms/step - loss: 0.2778 -
val_loss: 0.2541
Epoch 7/100
4/4 [==============================] - 0s 10ms/step - loss: 0.2276 -
val_loss: 0.2164
Epoch 8/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1935 -
val_loss: 0.1911
Epoch 9/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1699 -
val_loss: 0.1705
Epoch 10/100
4/4 [==============================] - 0s 11ms/step - loss: 0.1497 -
val_loss: 0.1527
Epoch 11/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1336 -
val_loss: 0.1397
Epoch 12/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1224 -
val_loss: 0.1292
Epoch 13/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1133 -
val_loss: 0.1204
Epoch 14/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1059 -
val_loss: 0.1135
Epoch 15/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0998 -
val_loss: 0.1072
Epoch 16/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0942 -
val_loss: 0.1017
Epoch 17/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0892 -
val_loss: 0.0971
Epoch 18/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0848 -
val_loss: 0.0931
Epoch 19/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0810 -
val_loss: 0.0895
Epoch 20/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0775 -
```

```
val_loss: 0.0860
Epoch 21/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0743 -
val_loss: 0.0827
Epoch 22/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0714 -
val_loss: 0.0798
Epoch 23/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0688 -
val_loss: 0.0771
Epoch 24/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0664 -
val_loss: 0.0746
Epoch 25/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0641 -
val_loss: 0.0724
Epoch 26/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0620 -
val_loss: 0.0702
Epoch 27/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0601 -
val_loss: 0.0681
Epoch 28/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0582 -
val_loss: 0.0662
Epoch 29/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0565 -
val_loss: 0.0644
Epoch 30/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0548 -
val_loss: 0.0627
Epoch 31/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0533 -
val_loss: 0.0610
Epoch 32/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0518 -
val_loss: 0.0594
Epoch 33/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0503 -
val_loss: 0.0579
Epoch 34/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0490 -
val_loss: 0.0564
Epoch 35/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0476 -
val_loss: 0.0549
Epoch 36/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0464 -
val_loss: 0.0535
```

```
Epoch 37/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0451 -
val_loss: 0.0522
Epoch 38/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0439 -
val_loss: 0.0508
Epoch 39/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0428 -
val_loss: 0.0495
Epoch 40/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0417 -
val_loss: 0.0483
Epoch 41/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0406 -
val_loss: 0.0471
Epoch 42/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0395 -
val_loss: 0.0459
Epoch 43/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0385 -
val_loss: 0.0448
Epoch 44/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0375 -
val_loss: 0.0437
Epoch 45/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0366 -
val_loss: 0.0426
Epoch 46/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0356 -
val_loss: 0.0416
Epoch 47/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0347 -
val_loss: 0.0406
Epoch 48/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0339 -
val_loss: 0.0397
Epoch 49/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0330 -
val_loss: 0.0387
Epoch 50/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0322 -
val_loss: 0.0378
Epoch 51/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0314 -
val_loss: 0.0370
Epoch 52/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0307 -
val_loss: 0.0361
Epoch 53/100
```

```
4/4 [==============================] - 0s 10ms/step - loss: 0.0300 -
val_loss: 0.0354
Epoch 54/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0293 -
val_loss: 0.0346
Epoch 55/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0287 -
val_loss: 0.0339
Epoch 56/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0280 -
val_loss: 0.0332
Epoch 57/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0274 -
val_loss: 0.0325
Epoch 58/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0268 -
val_loss: 0.0318
Epoch 59/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0262 -
val_loss: 0.0312
Epoch 60/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0257 -
val_loss: 0.0306
Epoch 61/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0251 -
val_loss: 0.0299
Epoch 62/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0246 -
val_loss: 0.0294
Epoch 63/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0240 -
val_loss: 0.0288
Epoch 64/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0235 -
val_loss: 0.0282
Epoch 65/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0230 -
val_loss: 0.0277
Epoch 66/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0226 -
val_loss: 0.0271
Epoch 67/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0221 -
val_loss: 0.0266
Epoch 68/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0217 -
val_loss: 0.0261
Epoch 69/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0213 -
val_loss: 0.0256
```

```
Epoch 70/100
4/4 [==============================] - 0s 16ms/step - loss: 0.0208 -
val_loss: 0.0252
Epoch 71/100
4/4 [==============================] - 0s 13ms/step - loss: 0.0204 -
val_loss: 0.0247
Epoch 72/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0201 -
val_loss: 0.0243
Epoch 73/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0197 -
val_loss: 0.0239
Epoch 74/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0193 -
val_loss: 0.0235
Epoch 75/100
4/4 [==============================] - 0s 13ms/step - loss: 0.0190 -
val_loss: 0.0231
Epoch 76/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0186 -
val_loss: 0.0227
Epoch 77/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0183 -
val_loss: 0.0224
Epoch 78/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0180 -
val_loss: 0.0220
Epoch 79/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0177 -
val_loss: 0.0217
Epoch 80/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0174 -
val_loss: 0.0214
Epoch 81/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0171 -
val_loss: 0.0211
Epoch 82/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0168 -
val_loss: 0.0208
Epoch 83/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0165 -
val_loss: 0.0205
Epoch 84/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0163 -
val_loss: 0.0202
Epoch 85/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0160 -
val_loss: 0.0199
Epoch 86/100
```

```
4/4 [==============================] - 0s 10ms/step - loss: 0.0158 -
val_loss: 0.0197
Epoch 87/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0155 -
val_loss: 0.0194
Epoch 88/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0153 -
val_loss: 0.0192
Epoch 89/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0151 -
val_loss: 0.0189
Epoch 90/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0149 -
val_loss: 0.0187
Epoch 91/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0147 -
val_loss: 0.0185
Epoch 92/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0145 -
val_loss: 0.0183
Epoch 93/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0143 -
val_loss: 0.0180
Epoch 94/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0141 -
val_loss: 0.0178
Epoch 95/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0139 -
val_loss: 0.0176
Epoch 96/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0137 -
val_loss: 0.0174
Epoch 97/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0135 -
val_loss: 0.0172
Epoch 98/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0134 -
val_loss: 0.0170
Epoch 99/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0132 -
val_loss: 0.0168
Epoch 100/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0130 -
val_loss: 0.0167

 Processing Batch 3/4
Epoch 1/100
4/4 [==============================] - 0s 37ms/step - loss: 1.8314 -
val_loss: 1.0910
```

```
Epoch 2/100
4/4 [==============================] - 0s 10ms/step - loss: 0.9840 -
val_loss: 0.6932
Epoch 3/100
4/4 [==============================] - 0s 11ms/step - loss: 0.6340 -
val_loss: 0.5082
Epoch 4/100
4/4 [==============================] - 0s 10ms/step - loss: 0.4653 -
val_loss: 0.4017
Epoch 5/100
4/4 [==============================] - 0s 10ms/step - loss: 0.3656 -
val_loss: 0.3296
Epoch 6/100
4/4 [==============================] - 0s 9ms/step - loss: 0.2988 -
val_loss: 0.2743
Epoch 7/100
4/4 [==============================] - 0s 10ms/step - loss: 0.2484 -
val_loss: 0.2328
Epoch 8/100
4/4 [==============================] - 0s 10ms/step - loss: 0.2137 -
val_loss: 0.2056
Epoch 9/100
4/4 [==============================] - 0s 9ms/step - loss: 0.1902 -
val_loss: 0.1848
Epoch 10/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1702 -
val_loss: 0.1675
Epoch 11/100
4/4 [==============================] - 0s 9ms/step - loss: 0.1533 -
val_loss: 0.1547
Epoch 12/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1403 -
val_loss: 0.1435
Epoch 13/100
4/4 [==============================] - 0s 9ms/step - loss: 0.1290 -
val_loss: 0.1332
Epoch 14/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1195 -
val_loss: 0.1252
Epoch 15/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1121 -
val_loss: 0.1187
Epoch 16/100
4/4 [==============================] - 0s 10ms/step - loss: 0.1058 -
val_loss: 0.1130
Epoch 17/100
4/4 [==============================] - 0s 11ms/step - loss: 0.1000 -
val_loss: 0.1074
Epoch 18/100
```

```
4/4 [==============================] - 0s 10ms/step - loss: 0.0948 -
val_loss: 0.1023
Epoch 19/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0901 -
val_loss: 0.0978
Epoch 20/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0860 -
val_loss: 0.0938
Epoch 21/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0824 -
val_loss: 0.0902
Epoch 22/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0791 -
val_loss: 0.0868
Epoch 23/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0762 -
val_loss: 0.0837
Epoch 24/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0735 -
val_loss: 0.0809
Epoch 25/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0710 -
val_loss: 0.0781
Epoch 26/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0686 -
val_loss: 0.0754
Epoch 27/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0663 -
val_loss: 0.0728
Epoch 28/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0641 -
val_loss: 0.0704
Epoch 29/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0619 -
val_loss: 0.0680
Epoch 30/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0599 -
val_loss: 0.0658
Epoch 31/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0579 -
val_loss: 0.0636
Epoch 32/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0561 -
val_loss: 0.0616
Epoch 33/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0544 -
val_loss: 0.0596
Epoch 34/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0527 -
```

```
val_loss: 0.0578
Epoch 35/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0512 -
val_loss: 0.0561
Epoch 36/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0497 -
val_loss: 0.0546
Epoch 37/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0484 -
val_loss: 0.0530
Epoch 38/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0471 -
val_loss: 0.0516
Epoch 39/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0458 -
val_loss: 0.0502
Epoch 40/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0446 -
val_loss: 0.0489
Epoch 41/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0434 -
val_loss: 0.0476
Epoch 42/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0423 -
val_loss: 0.0463
Epoch 43/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0412 -
val_loss: 0.0451
Epoch 44/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0401 -
val_loss: 0.0440
Epoch 45/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0391 -
val_loss: 0.0428
Epoch 46/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0381 -
val_loss: 0.0418
Epoch 47/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0371 -
val_loss: 0.0407
Epoch 48/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0362 -
val_loss: 0.0397
Epoch 49/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0353 -
val_loss: 0.0388
Epoch 50/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0345 -
val_loss: 0.0378
```

```
Epoch 51/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0336 -
val_loss: 0.0370
Epoch 52/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0328 -
val_loss: 0.0361
Epoch 53/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0321 -
val_loss: 0.0353
Epoch 54/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0313 -
val_loss: 0.0345
Epoch 55/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0306 -
val_loss: 0.0338
Epoch 56/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0299 -
val_loss: 0.0331
Epoch 57/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0292 -
val_loss: 0.0324
Epoch 58/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0286 -
val_loss: 0.0317
Epoch 59/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0280 -
val_loss: 0.0311
Epoch 60/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0274 -
val_loss: 0.0305
Epoch 61/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0268 -
val_loss: 0.0299
Epoch 62/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0263 -
val_loss: 0.0293
Epoch 63/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0257 -
val_loss: 0.0288
Epoch 64/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0252 -
val_loss: 0.0282
Epoch 65/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0247 -
val_loss: 0.0277
Epoch 66/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0242 -
val_loss: 0.0272
Epoch 67/100
```

```
4/4 [==============================] - 0s 11ms/step - loss: 0.0238 -
val_loss: 0.0268
Epoch 68/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0233 -
val_loss: 0.0263
Epoch 69/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0229 -
val_loss: 0.0258
Epoch 70/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0224 -
val_loss: 0.0253
Epoch 71/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0220 -
val_loss: 0.0248
Epoch 72/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0216 -
val_loss: 0.0244
Epoch 73/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0212 -
val_loss: 0.0240
Epoch 74/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0208 -
val_loss: 0.0236
Epoch 75/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0205 -
val_loss: 0.0232
Epoch 76/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0201 -
val_loss: 0.0228
Epoch 77/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0197 -
val_loss: 0.0224
Epoch 78/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0194 -
val_loss: 0.0221
Epoch 79/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0191 -
val_loss: 0.0217
Epoch 80/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0188 -
val_loss: 0.0213
Epoch 81/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0185 -
val_loss: 0.0210
Epoch 82/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0182 -
val_loss: 0.0207
Epoch 83/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0179 -
```

```
val_loss: 0.0203
Epoch 84/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0176 -
val_loss: 0.0200
Epoch 85/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0173 -
val_loss: 0.0197
Epoch 86/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0170 -
val_loss: 0.0194
Epoch 87/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0168 -
val_loss: 0.0191
Epoch 88/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0165 -
val_loss: 0.0189
Epoch 89/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0163 -
val_loss: 0.0186
Epoch 90/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0160 -
val_loss: 0.0183
Epoch 91/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0158 -
val_loss: 0.0180
Epoch 92/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0156 -
val_loss: 0.0178
Epoch 93/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0153 -
val_loss: 0.0175
Epoch 94/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0151 -
val_loss: 0.0173
Epoch 95/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0149 -
val_loss: 0.0170
Epoch 96/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0147 -
val_loss: 0.0168
Epoch 97/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0145 -
val_loss: 0.0166
Epoch 98/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0143 -
val_loss: 0.0164
Epoch 99/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0141 -
val_loss: 0.0161
```

```
Epoch 100/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0139 -
val_loss: 0.0159

 Processing Batch 4/4
Epoch 1/100
2/2 [==============================] - 1s 288ms/step - loss: 2.2388 -
val_loss: 1.5020
Epoch 2/100
2/2 [==============================] - 0s 21ms/step - loss: 1.5618 -
val_loss: 1.1037
Epoch 3/100
2/2 [==============================] - 0s 20ms/step - loss: 1.1481 -
val_loss: 0.8445
Epoch 4/100
2/2 [==============================] - 0s 20ms/step - loss: 0.8790 -
val_loss: 0.6712
Epoch 5/100
2/2 [==============================] - 0s 22ms/step - loss: 0.7030 -
val_loss: 0.5539
Epoch 6/100
2/2 [==============================] - 0s 20ms/step - loss: 0.5819 -
val_loss: 0.4674
Epoch 7/100
2/2 [==============================] - 0s 20ms/step - loss: 0.4928 -
val_loss: 0.4019
Epoch 8/100
2/2 [==============================] - 0s 22ms/step - loss: 0.4246 -
val_loss: 0.3507
Epoch 9/100
2/2 [==============================] - 0s 19ms/step - loss: 0.3718 -
val_loss: 0.3088
Epoch 10/100
2/2 [==============================] - 0s 25ms/step - loss: 0.3293 -
val_loss: 0.2740
Epoch 11/100
2/2 [==============================] - 0s 21ms/step - loss: 0.2944 -
val_loss: 0.2463
Epoch 12/100
2/2 [==============================] - 0s 19ms/step - loss: 0.2654 -
val_loss: 0.2234
Epoch 13/100
2/2 [==============================] - 0s 20ms/step - loss: 0.2406 -
val_loss: 0.2052
Epoch 14/100
2/2 [==============================] - 0s 21ms/step - loss: 0.2201 -
val_loss: 0.1914
Epoch 15/100
2/2 [==============================] - 0s 20ms/step - loss: 0.2042 -
```

```
val_loss: 0.1806
Epoch 16/100
2/2 [==============================] - 0s 19ms/step - loss: 0.1913 -
val_loss: 0.1717
Epoch 17/100
2/2 [==============================] - 0s 21ms/step - loss: 0.1805 -
val_loss: 0.1637
Epoch 18/100
2/2 [==============================] - 0s 19ms/step - loss: 0.1707 -
val_loss: 0.1550
Epoch 19/100
2/2 [==============================] - 0s 20ms/step - loss: 0.1605 -
val_loss: 0.1455
Epoch 20/100
2/2 [==============================] - 0s 25ms/step - loss: 0.1502 -
val_loss: 0.1360
Epoch 21/100
2/2 [==============================] - 0s 20ms/step - loss: 0.1402 -
val_loss: 0.1271
Epoch 22/100
2/2 [==============================] - 0s 22ms/step - loss: 0.1311 -
val_loss: 0.1195
Epoch 23/100
2/2 [==============================] - 0s 19ms/step - loss: 0.1235 -
val_loss: 0.1132
Epoch 24/100
2/2 [==============================] - 0s 19ms/step - loss: 0.1171 -
val_loss: 0.1079
Epoch 25/100
2/2 [==============================] - 0s 21ms/step - loss: 0.1117 -
val_loss: 0.1033
Epoch 26/100
2/2 [==============================] - 0s 20ms/step - loss: 0.1068 -
val_loss: 0.0993
Epoch 27/100
2/2 [==============================] - 0s 20ms/step - loss: 0.1024 -
val_loss: 0.0959
Epoch 28/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0987 -
val_loss: 0.0931
Epoch 29/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0955 -
val_loss: 0.0907
Epoch 30/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0927 -
val_loss: 0.0882
Epoch 31/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0899 -
val_loss: 0.0858
```

```
Epoch 32/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0873 -
val_loss: 0.0834
Epoch 33/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0847 -
val_loss: 0.0811
Epoch 34/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0823 -
val_loss: 0.0790
Epoch 35/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0800 -
val_loss: 0.0771
Epoch 36/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0778 -
val_loss: 0.0753
Epoch 37/100
2/2 [==============================] - 0s 18ms/step - loss: 0.0758 -
val_loss: 0.0736
Epoch 38/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0738 -
val_loss: 0.0719
Epoch 39/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0720 -
val_loss: 0.0704
Epoch 40/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0703 -
val_loss: 0.0689
Epoch 41/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0687 -
val_loss: 0.0675
Epoch 42/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0673 -
val_loss: 0.0663
Epoch 43/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0660 -
val_loss: 0.0650
Epoch 44/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0647 -
val_loss: 0.0638
Epoch 45/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0635 -
val_loss: 0.0627
Epoch 46/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0623 -
val_loss: 0.0617
Epoch 47/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0612 -
val_loss: 0.0607
Epoch 48/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0602 -
```

```
val_loss: 0.0597
Epoch 49/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0592 -
val_loss: 0.0588
Epoch 50/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0583 -
val_loss: 0.0579
Epoch 51/100
2/2 [==============================] - 0s 19ms/step - loss: 0.0574 -
val_loss: 0.0569
Epoch 52/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0565 -
val_loss: 0.0560
Epoch 53/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0557 -
val_loss: 0.0551
Epoch 54/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0548 -
val_loss: 0.0543
Epoch 55/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0540 -
val_loss: 0.0535
Epoch 56/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0533 -
val_loss: 0.0527
Epoch 57/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0525 -
val_loss: 0.0519
Epoch 58/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0518 -
val_loss: 0.0512
Epoch 59/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0511 -
val_loss: 0.0505
Epoch 60/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0504 -
val_loss: 0.0498
Epoch 61/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0497 -
val_loss: 0.0491
Epoch 62/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0490 -
val_loss: 0.0484
Epoch 63/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0484 -
val_loss: 0.0478
Epoch 64/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0477 -
val_loss: 0.0471
```

```
Epoch 65/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0471 -
val_loss: 0.0465
Epoch 66/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0465 -
val_loss: 0.0459
Epoch 67/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0459 -
val_loss: 0.0453
Epoch 68/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0453 -
val_loss: 0.0447
Epoch 69/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0447 -
val_loss: 0.0442
Epoch 70/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0441 -
val_loss: 0.0437
Epoch 71/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0436 -
val_loss: 0.0431
Epoch 72/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0431 -
val_loss: 0.0426
Epoch 73/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0425 -
val_loss: 0.0422
Epoch 74/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0420 -
val_loss: 0.0417
Epoch 75/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0415 -
val_loss: 0.0412
Epoch 76/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0410 -
val_loss: 0.0408
Epoch 77/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0405 -
val_loss: 0.0403
Epoch 78/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0400 -
val_loss: 0.0399
Epoch 79/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0396 -
val_loss: 0.0394
Epoch 80/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0391 -
val_loss: 0.0390
Epoch 81/100
```

```
2/2 [==============================] - 0s 23ms/step - loss: 0.0386 -
val_loss: 0.0385
Epoch 82/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0381 -
val_loss: 0.0381
Epoch 83/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0377 -
val_loss: 0.0376
Epoch 84/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0372 -
val_loss: 0.0372
Epoch 85/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0368 -
val_loss: 0.0368
Epoch 86/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0364 -
val_loss: 0.0363
Epoch 87/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0359 -
val_loss: 0.0359
Epoch 88/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0355 -
val_loss: 0.0355
Epoch 89/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0351 -
val_loss: 0.0351
Epoch 90/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0347 -
val_loss: 0.0348
Epoch 91/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0343 -
val_loss: 0.0344
Epoch 92/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0339 -
val_loss: 0.0340
Epoch 93/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0336 -
val_loss: 0.0336
Epoch 94/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0332 -
val_loss: 0.0333
Epoch 95/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0328 -
val_loss: 0.0329
Epoch 96/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0324 -
val_loss: 0.0326
Epoch 97/100
2/2 [==============================] - 0s 27ms/step - loss: 0.0321 -
```

```
val_loss: 0.0322
Epoch 98/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0317 -
val_loss: 0.0319
Epoch 99/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0314 -
val_loss: 0.0316
Epoch 100/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0310 -
val_loss: 0.0312

Results:
              precision    recall  f1-score   support

           0       1.00      0.98      0.99      1670
           1       0.20      0.80      0.31        10

    accuracy                           0.98      1680
   macro avg       0.60      0.89      0.65      1680
weighted avg       0.99      0.98      0.99      1680

Precision: 0.1951
Recall:    0.8000
F1 Score:  0.3137
ROC-AUC:   0.9764
runtime: 0.2699 minutes
```

AE STREAMING VARIANT 2

```
#Parametes
window_size = 20
rolling_window = 10
batch_size = 500
max_epochs = 100

data_raw = result['measurement'].values
labels = result['anomaly'].values
n_samples = len(data_raw)

all_preds = np.zeros(n_samples)
all_mse = np.zeros(n_samples)

initializer = initializers.HeNormal(seed=seed)

def create_windows(data, window_size):
    X = []
    for i in range(len(data) - window_size + 1):
```

```
        X.append(data[i:i+window_size])
    return np.array(X)

num_batches = n_samples // batch_size + (1 if n_samples % batch_size !
= 0 else 0)

model = None


start_time = time.time()

for b in range(num_batches):
    print(f"\nProcessing Batch {b+1}/{num_batches}")

    start_idx = b * batch_size
    end_idx = min(start_idx + batch_size, n_samples)
    batch_raw = data_raw[start_idx:end_idx]
    batch_labels = labels[start_idx:end_idx]

    #Rolling mean & std
    rolling_mean = pd.Series(batch_raw).rolling(window=rolling_window,
min_periods=1).mean().values
    rolling_std = pd.Series(batch_raw).rolling(window=rolling_window,
min_periods=1).std().fillna(0).values

    #Features
    batch_features = np.column_stack([batch_raw, rolling_mean,
rolling_std])

    #Scaling
    scaler = StandardScaler()
    batch_features_scaled = scaler.fit_transform(batch_features)

    #Sliding windows
    if len(batch_features_scaled) < window_size:
        print("Skip batch: not enough data for sliding windows")
        continue

    X_batch_all = create_windows(batch_features_scaled, window_size)
    y_batch_true = batch_labels[window_size-1:]

    #Train using normal only
    X_batch_train = X_batch_all[y_batch_true == 0]

    if len(X_batch_train) < 10:
        print("Skip batch: too few normal samples for training")
        continue

    input_shape = (window_size, X_batch_all.shape[2])

    if model is None:
```

```
        model = models.Sequential([
            layers.Input(shape=input_shape),
            layers.Conv1D(32, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
            layers.Conv1D(16, kernel_size=3, activation='relu',
padding='same', kernel_initializer=initializer),
            layers.Conv1DTranspose(16, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
            layers.Conv1DTranspose(32, kernel_size=3, padding='same',
activation='relu', kernel_initializer=initializer),
            layers.Conv1D(X_batch_all.shape[2], kernel_size=1,
activation='linear', padding='same', kernel_initializer=initializer)
        ])
        model.compile(optimizer='adam', loss='mse')


    X_train_train, X_val = train_test_split(X_batch_train,
test_size=0.1, random_state=seed)
    early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True, verbose=0)

    model.fit(
        X_train_train, X_train_train,
        validation_data=(X_val, X_val),
        epochs=max_epochs,
        batch_size=128,
        shuffle=False,
        callbacks=[early_stop],
        verbose=1
    )


    X_pred = model.predict(X_batch_all, verbose=0)
    mse = np.mean(np.square(X_batch_all - X_pred), axis=(1, 2))

    #Threshold search
    best_f1 = 0
    best_thresh = np.percentile(mse, 90)

    for p in range(80, 100):
        threshold = np.percentile(mse, p)
        y_pred = (mse > threshold).astype(int)
        f1 = f1_score(y_batch_true, y_pred)
        if f1 > best_f1:
            best_f1 = f1
            best_thresh = threshold

    y_batch_pred = (mse > best_thresh).astype(int)
```

```
    batch_pred_full = np.zeros(end_idx - start_idx)
    batch_mse_full = np.zeros(end_idx - start_idx)

    batch_pred_full[window_size-1:] = y_batch_pred
    batch_mse_full[window_size-1:] = mse

    all_preds[start_idx:end_idx] = batch_pred_full
    all_mse[start_idx:end_idx] = batch_mse_full



end_time = time.time()
elapsed_time = end_time - start_time


result['prediction'] = all_preds.astype(int)
result['reconstruction_error'] = all_mse

print("\nResults:")
print(classification_report(labels, all_preds.astype(int)))

print(f"Precision: {precision_score(labels,
all_preds.astype(int)):.4f}")
print(f"Recall:    {recall_score(labels, all_preds.astype(int)):.4f}")
print(f"F1 Score:  {f1_score(labels, all_preds.astype(int)):.4f}")
print(f"ROC-AUC:   {roc_auc_score(labels, all_mse):.4f}")
print(f"runtime: {elapsed_time / 60:.4f} minutes")


Processing Batch 1/4
Epoch 1/100
4/4 [==============================] - 0s 58ms/step - loss: 1.8839 -
val_loss: 1.1209
Epoch 2/100
4/4 [==============================] - 0s 30ms/step - loss: 1.0199 -
val_loss: 0.7014
Epoch 3/100
4/4 [==============================] - 0s 32ms/step - loss: 0.6617 -
val_loss: 0.5157
Epoch 4/100
4/4 [==============================] - 0s 8ms/step - loss: 0.4910 -
val_loss: 0.4061
Epoch 5/100
4/4 [==============================] - 0s 9ms/step - loss: 0.3879 -
val_loss: 0.3292
Epoch 6/100
4/4 [==============================] - 0s 8ms/step - loss: 0.3185 -
val_loss: 0.2717
Epoch 7/100
4/4 [==============================] - 0s 8ms/step - loss: 0.2676 -
```

```
val_loss: 0.2283
Epoch 8/100
4/4 [==============================] - 0s 9ms/step - loss: 0.2315 -
val_loss: 0.1993
Epoch 9/100
4/4 [==============================] - 0s 9ms/step - loss: 0.2075 -
val_loss: 0.1779
Epoch 10/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1884 -
val_loss: 0.1615
Epoch 11/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1732 -
val_loss: 0.1495
Epoch 12/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1611 -
val_loss: 0.1389
Epoch 13/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1493 -
val_loss: 0.1290
Epoch 14/100
4/4 [==============================] - 0s 9ms/step - loss: 0.1386 -
val_loss: 0.1212
Epoch 15/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1302 -
val_loss: 0.1151
Epoch 16/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1231 -
val_loss: 0.1098
Epoch 17/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1167 -
val_loss: 0.1047
Epoch 18/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1107 -
val_loss: 0.0997
Epoch 19/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1052 -
val_loss: 0.0950
Epoch 20/100
4/4 [==============================] - 0s 8ms/step - loss: 0.1003 -
val_loss: 0.0908
Epoch 21/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0958 -
val_loss: 0.0872
Epoch 22/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0918 -
val_loss: 0.0840
Epoch 23/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0881 -
val_loss: 0.0809
```

```
Epoch 24/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0848 -
val_loss: 0.0780
Epoch 25/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0817 -
val_loss: 0.0754
Epoch 26/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0788 -
val_loss: 0.0729
Epoch 27/100
4/4 [==============================] - 0s 7ms/step - loss: 0.0762 -
val_loss: 0.0707
Epoch 28/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0737 -
val_loss: 0.0686
Epoch 29/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0714 -
val_loss: 0.0666
Epoch 30/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0691 -
val_loss: 0.0647
Epoch 31/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0670 -
val_loss: 0.0629
Epoch 32/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0651 -
val_loss: 0.0612
Epoch 33/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0632 -
val_loss: 0.0597
Epoch 34/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0615 -
val_loss: 0.0582
Epoch 35/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0598 -
val_loss: 0.0567
Epoch 36/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0582 -
val_loss: 0.0553
Epoch 37/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0566 -
val_loss: 0.0540
Epoch 38/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0551 -
val_loss: 0.0526
Epoch 39/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0537 -
val_loss: 0.0514
Epoch 40/100
```

```
4/4 [==============================] - 0s 8ms/step - loss: 0.0523 -
val_loss: 0.0501
Epoch 41/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0509 -
val_loss: 0.0489
Epoch 42/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0496 -
val_loss: 0.0478
Epoch 43/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0483 -
val_loss: 0.0466
Epoch 44/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0471 -
val_loss: 0.0456
Epoch 45/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0458 -
val_loss: 0.0445
Epoch 46/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0447 -
val_loss: 0.0434
Epoch 47/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0435 -
val_loss: 0.0424
Epoch 48/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0424 -
val_loss: 0.0414
Epoch 49/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0413 -
val_loss: 0.0404
Epoch 50/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0402 -
val_loss: 0.0395
Epoch 51/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0392 -
val_loss: 0.0385
Epoch 52/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0382 -
val_loss: 0.0376
Epoch 53/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0372 -
val_loss: 0.0368
Epoch 54/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0363 -
val_loss: 0.0359
Epoch 55/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0354 -
val_loss: 0.0351
Epoch 56/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0345 -
```

```
val_loss: 0.0343
Epoch 57/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0337 -
val_loss: 0.0335
Epoch 58/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0329 -
val_loss: 0.0328
Epoch 59/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0321 -
val_loss: 0.0320
Epoch 60/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0313 -
val_loss: 0.0313
Epoch 61/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0306 -
val_loss: 0.0306
Epoch 62/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0298 -
val_loss: 0.0299
Epoch 63/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0291 -
val_loss: 0.0293
Epoch 64/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0285 -
val_loss: 0.0287
Epoch 65/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0278 -
val_loss: 0.0281
Epoch 66/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0272 -
val_loss: 0.0276
Epoch 67/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0266 -
val_loss: 0.0270
Epoch 68/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0261 -
val_loss: 0.0266
Epoch 69/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0255 -
val_loss: 0.0261
Epoch 70/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0250 -
val_loss: 0.0256
Epoch 71/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0245 -
val_loss: 0.0252
Epoch 72/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0241 -
val_loss: 0.0248
```

```
Epoch 73/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0237 -
val_loss: 0.0245
Epoch 74/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0232 -
val_loss: 0.0241
Epoch 75/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0228 -
val_loss: 0.0237
Epoch 76/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0224 -
val_loss: 0.0233
Epoch 77/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0221 -
val_loss: 0.0230
Epoch 78/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0217 -
val_loss: 0.0227
Epoch 79/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0214 -
val_loss: 0.0223
Epoch 80/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0210 -
val_loss: 0.0220
Epoch 81/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0207 -
val_loss: 0.0217
Epoch 82/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0204 -
val_loss: 0.0214
Epoch 83/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0201 -
val_loss: 0.0211
Epoch 84/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0198 -
val_loss: 0.0208
Epoch 85/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0195 -
val_loss: 0.0206
Epoch 86/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0192 -
val_loss: 0.0203
Epoch 87/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0190 -
val_loss: 0.0200
Epoch 88/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0187 -
val_loss: 0.0198
Epoch 89/100
```

```
4/4 [==============================] - 0s 8ms/step - loss: 0.0184 -
val_loss: 0.0195
Epoch 90/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0182 -
val_loss: 0.0193
Epoch 91/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0179 -
val_loss: 0.0190
Epoch 92/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0177 -
val_loss: 0.0188
Epoch 93/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0175 -
val_loss: 0.0186
Epoch 94/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0173 -
val_loss: 0.0184
Epoch 95/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0170 -
val_loss: 0.0181
Epoch 96/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0168 -
val_loss: 0.0179
Epoch 97/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0166 -
val_loss: 0.0177
Epoch 98/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0164 -
val_loss: 0.0175
Epoch 99/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0162 -
val_loss: 0.0173
Epoch 100/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0160 -
val_loss: 0.0171

Processing Batch 2/4
Epoch 1/100
4/4 [==============================] - 0s 15ms/step - loss: 0.0188 -
val_loss: 0.0212
Epoch 2/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0182 -
val_loss: 0.0205
Epoch 3/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0176 -
val_loss: 0.0200
Epoch 4/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0169 -
val_loss: 0.0195
```

```
Epoch 5/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0164 -
val_loss: 0.0191
Epoch 6/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0159 -
val_loss: 0.0187
Epoch 7/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0154 -
val_loss: 0.0183
Epoch 8/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0150 -
val_loss: 0.0179
Epoch 9/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0146 -
val_loss: 0.0176
Epoch 10/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0143 -
val_loss: 0.0173
Epoch 11/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0140 -
val_loss: 0.0170
Epoch 12/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0137 -
val_loss: 0.0168
Epoch 13/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0134 -
val_loss: 0.0166
Epoch 14/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0132 -
val_loss: 0.0163
Epoch 15/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0130 -
val_loss: 0.0161
Epoch 16/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0127 -
val_loss: 0.0159
Epoch 17/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0125 -
val_loss: 0.0157
Epoch 18/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0123 -
val_loss: 0.0155
Epoch 19/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0122 -
val_loss: 0.0153
Epoch 20/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0120 -
val_loss: 0.0151
Epoch 21/100
```

```
4/4 [==============================] - 0s 9ms/step - loss: 0.0118 -
val_loss: 0.0149
Epoch 22/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0116 -
val_loss: 0.0147
Epoch 23/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0115 -
val_loss: 0.0145
Epoch 24/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0113 -
val_loss: 0.0144
Epoch 25/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0112 -
val_loss: 0.0142
Epoch 26/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0110 -
val_loss: 0.0140
Epoch 27/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0109 -
val_loss: 0.0139
Epoch 28/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0107 -
val_loss: 0.0137
Epoch 29/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0106 -
val_loss: 0.0136
Epoch 30/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0105 -
val_loss: 0.0135
Epoch 31/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0103 -
val_loss: 0.0133
Epoch 32/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0102 -
val_loss: 0.0132
Epoch 33/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0101 -
val_loss: 0.0131
Epoch 34/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0100 -
val_loss: 0.0129
Epoch 35/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0099 -
val_loss: 0.0128
Epoch 36/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0098 -
val_loss: 0.0127
Epoch 37/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0096 -
```

```
val_loss: 0.0126
Epoch 38/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0095 -
val_loss: 0.0124
Epoch 39/100
4/4 [==============================] - 0s 8ms/step - loss: 0.0094 -
val_loss: 0.0123
Epoch 40/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0093 -
val_loss: 0.0122
Epoch 41/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0092 -
val_loss: 0.0121
Epoch 42/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0091 -
val_loss: 0.0120
Epoch 43/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0090 -
val_loss: 0.0119
Epoch 44/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0090 -
val_loss: 0.0118
Epoch 45/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0089 -
val_loss: 0.0117
Epoch 46/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0088 -
val_loss: 0.0116
Epoch 47/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0087 -
val_loss: 0.0115
Epoch 48/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0086 -
val_loss: 0.0114
Epoch 49/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0085 -
val_loss: 0.0113
Epoch 50/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0085 -
val_loss: 0.0112
Epoch 51/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0084 -
val_loss: 0.0111
Epoch 52/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0083 -
val_loss: 0.0110
Epoch 53/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0082 -
val_loss: 0.0109
```

```
Epoch 54/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0081 -
val_loss: 0.0109
Epoch 55/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0081 -
val_loss: 0.0108
Epoch 56/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0080 -
val_loss: 0.0107
Epoch 57/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0079 -
val_loss: 0.0106
Epoch 58/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0079 -
val_loss: 0.0105
Epoch 59/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0078 -
val_loss: 0.0104
Epoch 60/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0077 -
val_loss: 0.0103
Epoch 61/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0077 -
val_loss: 0.0103
Epoch 62/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0076 -
val_loss: 0.0102
Epoch 63/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0075 -
val_loss: 0.0101
Epoch 64/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0075 -
val_loss: 0.0100
Epoch 65/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0074 -
val_loss: 0.0100
Epoch 66/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0074 -
val_loss: 0.0099
Epoch 67/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0073 -
val_loss: 0.0098
Epoch 68/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0072 -
val_loss: 0.0098
Epoch 69/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0072 -
val_loss: 0.0097
Epoch 70/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0071 -
```

```
val_loss: 0.0096
Epoch 71/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0071 -
val_loss: 0.0095
Epoch 72/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0070 -
val_loss: 0.0095
Epoch 73/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0070 -
val_loss: 0.0094
Epoch 74/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0069 -
val_loss: 0.0093
Epoch 75/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0069 -
val_loss: 0.0093
Epoch 76/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0068 -
val_loss: 0.0092
Epoch 77/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0068 -
val_loss: 0.0091
Epoch 78/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0067 -
val_loss: 0.0091
Epoch 79/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0067 -
val_loss: 0.0090
Epoch 80/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0066 -
val_loss: 0.0090
Epoch 81/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0066 -
val_loss: 0.0089
Epoch 82/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0065 -
val_loss: 0.0088
Epoch 83/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0065 -
val_loss: 0.0088
Epoch 84/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0064 -
val_loss: 0.0087
Epoch 85/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0064 -
val_loss: 0.0087
Epoch 86/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0063 -
val_loss: 0.0086
```

```
Epoch 87/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0063 -
val_loss: 0.0085
Epoch 88/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0062 -
val_loss: 0.0085
Epoch 89/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0062 -
val_loss: 0.0084
Epoch 90/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0062 -
val_loss: 0.0084
Epoch 91/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0061 -
val_loss: 0.0083
Epoch 92/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0061 -
val_loss: 0.0083
Epoch 93/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0060 -
val_loss: 0.0082
Epoch 94/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0060 -
val_loss: 0.0082
Epoch 95/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0060 -
val_loss: 0.0081
Epoch 96/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0059 -
val_loss: 0.0081
Epoch 97/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0059 -
val_loss: 0.0080
Epoch 98/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0058 -
val_loss: 0.0080
Epoch 99/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0058 -
val_loss: 0.0079
Epoch 100/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0058 -
val_loss: 0.0079

Processing Batch 3/4
Epoch 1/100
4/4 [==============================] - 0s 17ms/step - loss: 0.0117 -
val_loss: 0.0134
Epoch 2/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0112 -
```

```
val_loss: 0.0128
Epoch 3/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0107 -
val_loss: 0.0121
Epoch 4/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0102 -
val_loss: 0.0116
Epoch 5/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0098 -
val_loss: 0.0111
Epoch 6/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0094 -
val_loss: 0.0108
Epoch 7/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0091 -
val_loss: 0.0105
Epoch 8/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0089 -
val_loss: 0.0102
Epoch 9/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0086 -
val_loss: 0.0100
Epoch 10/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0084 -
val_loss: 0.0098
Epoch 11/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0083 -
val_loss: 0.0096
Epoch 12/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0081 -
val_loss: 0.0094
Epoch 13/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0079 -
val_loss: 0.0092
Epoch 14/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0078 -
val_loss: 0.0091
Epoch 15/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0077 -
val_loss: 0.0089
Epoch 16/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0075 -
val_loss: 0.0088
Epoch 17/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0074 -
val_loss: 0.0087
Epoch 18/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0073 -
val_loss: 0.0086
```

```
Epoch 19/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0072 -
val_loss: 0.0084
Epoch 20/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0071 -
val_loss: 0.0083
Epoch 21/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0070 -
val_loss: 0.0082
Epoch 22/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0069 -
val_loss: 0.0081
Epoch 23/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0068 -
val_loss: 0.0080
Epoch 24/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0067 -
val_loss: 0.0079
Epoch 25/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0066 -
val_loss: 0.0078
Epoch 26/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0066 -
val_loss: 0.0078
Epoch 27/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0065 -
val_loss: 0.0077
Epoch 28/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0064 -
val_loss: 0.0076
Epoch 29/100
4/4 [==============================] - 0s 9ms/step - loss: 0.0063 -
val_loss: 0.0075
Epoch 30/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0063 -
val_loss: 0.0074
Epoch 31/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0062 -
val_loss: 0.0074
Epoch 32/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0061 -
val_loss: 0.0073
Epoch 33/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0061 -
val_loss: 0.0072
Epoch 34/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0060 -
val_loss: 0.0072
Epoch 35/100
```

```
4/4 [==============================] - 0s 10ms/step - loss: 0.0059 -
val_loss: 0.0071
Epoch 36/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0059 -
val_loss: 0.0070
Epoch 37/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0058 -
val_loss: 0.0070
Epoch 38/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0058 -
val_loss: 0.0069
Epoch 39/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0057 -
val_loss: 0.0069
Epoch 40/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0057 -
val_loss: 0.0068
Epoch 41/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0056 -
val_loss: 0.0068
Epoch 42/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0056 -
val_loss: 0.0067
Epoch 43/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0055 -
val_loss: 0.0066
Epoch 44/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0055 -
val_loss: 0.0066
Epoch 45/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0054 -
val_loss: 0.0065
Epoch 46/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0054 -
val_loss: 0.0065
Epoch 47/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0053 -
val_loss: 0.0065
Epoch 48/100
4/4 [==============================] - 0s 15ms/step - loss: 0.0053 -
val_loss: 0.0064
Epoch 49/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0052 -
val_loss: 0.0064
Epoch 50/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0052 -
val_loss: 0.0063
Epoch 51/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0052 -
```

```
val_loss: 0.0063
Epoch 52/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0051 -
val_loss: 0.0062
Epoch 53/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0051 -
val_loss: 0.0062
Epoch 54/100
4/4 [==============================] - 0s 12ms/step - loss: 0.0050 -
val_loss: 0.0061
Epoch 55/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0050 -
val_loss: 0.0061
Epoch 56/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0050 -
val_loss: 0.0061
Epoch 57/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0049 -
val_loss: 0.0060
Epoch 58/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0049 -
val_loss: 0.0060
Epoch 59/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0048 -
val_loss: 0.0059
Epoch 60/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0048 -
val_loss: 0.0059
Epoch 61/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0048 -
val_loss: 0.0059
Epoch 62/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0047 -
val_loss: 0.0058
Epoch 63/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0047 -
val_loss: 0.0058
Epoch 64/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0047 -
val_loss: 0.0057
Epoch 65/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0046 -
val_loss: 0.0057
Epoch 66/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0046 -
val_loss: 0.0057
Epoch 67/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0046 -
val_loss: 0.0056
```

```
Epoch 68/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0045 -
val_loss: 0.0056
Epoch 69/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0045 -
val_loss: 0.0056
Epoch 70/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0045 -
val_loss: 0.0055
Epoch 71/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0044 -
val_loss: 0.0055
Epoch 72/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0044 -
val_loss: 0.0055
Epoch 73/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0044 -
val_loss: 0.0054
Epoch 74/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0043 -
val_loss: 0.0054
Epoch 75/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0043 -
val_loss: 0.0054
Epoch 76/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0043 -
val_loss: 0.0053
Epoch 77/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0043 -
val_loss: 0.0053
Epoch 78/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0042 -
val_loss: 0.0053
Epoch 79/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0042 -
val_loss: 0.0052
Epoch 80/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0042 -
val_loss: 0.0052
Epoch 81/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0041 -
val_loss: 0.0052
Epoch 82/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0041 -
val_loss: 0.0051
Epoch 83/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0041 -
val_loss: 0.0051
Epoch 84/100
```

```
4/4 [==============================] - 0s 10ms/step - loss: 0.0041 -
val_loss: 0.0051
Epoch 85/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0040 -
val_loss: 0.0051
Epoch 86/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0040 -
val_loss: 0.0050
Epoch 87/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0040 -
val_loss: 0.0050
Epoch 88/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0040 -
val_loss: 0.0050
Epoch 89/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0039 -
val_loss: 0.0050
Epoch 90/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0039 -
val_loss: 0.0049
Epoch 91/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0039 -
val_loss: 0.0049
Epoch 92/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0039 -
val_loss: 0.0049
Epoch 93/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0039 -
val_loss: 0.0049
Epoch 94/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0038 -
val_loss: 0.0048
Epoch 95/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0038 -
val_loss: 0.0048
Epoch 96/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0038 -
val_loss: 0.0048
Epoch 97/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0038 -
val_loss: 0.0048
Epoch 98/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0037 -
val_loss: 0.0047
Epoch 99/100
4/4 [==============================] - 0s 10ms/step - loss: 0.0037 -
val_loss: 0.0047
Epoch 100/100
4/4 [==============================] - 0s 11ms/step - loss: 0.0037 -
```

```
val_loss: 0.0047

Processing Batch 4/4
Epoch 1/100
2/2 [==============================] - 0s 42ms/step - loss: 0.0047 -
val_loss: 0.0041
Epoch 2/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0045 -
val_loss: 0.0041
Epoch 3/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0043 -
val_loss: 0.0040
Epoch 4/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0042 -
val_loss: 0.0038
Epoch 5/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0040 -
val_loss: 0.0037
Epoch 6/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0039 -
val_loss: 0.0036
Epoch 7/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0038 -
val_loss: 0.0036
Epoch 8/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0037 -
val_loss: 0.0036
Epoch 9/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0037 -
val_loss: 0.0036
Epoch 10/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0036 -
val_loss: 0.0035
Epoch 11/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0035 -
val_loss: 0.0035
Epoch 12/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0035 -
val_loss: 0.0034
Epoch 13/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0034 -
val_loss: 0.0034
Epoch 14/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0034 -
val_loss: 0.0034
Epoch 15/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0033 -
val_loss: 0.0033
Epoch 16/100
```

```
2/2 [==============================] - 0s 21ms/step - loss: 0.0033 -
val_loss: 0.0033
Epoch 17/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0033 -
val_loss: 0.0032
Epoch 18/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0032 -
val_loss: 0.0032
Epoch 19/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0032 -
val_loss: 0.0032
Epoch 20/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0031 -
val_loss: 0.0032
Epoch 21/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0031 -
val_loss: 0.0031
Epoch 22/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0031 -
val_loss: 0.0031
Epoch 23/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0031 -
val_loss: 0.0031
Epoch 24/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0030 -
val_loss: 0.0031
Epoch 25/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0030 -
val_loss: 0.0031
Epoch 26/100
2/2 [==============================] - 0s 27ms/step - loss: 0.0030 -
val_loss: 0.0030
Epoch 27/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0030 -
val_loss: 0.0030
Epoch 28/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0029 -
val_loss: 0.0030
Epoch 29/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0029 -
val_loss: 0.0030
Epoch 30/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0029 -
val_loss: 0.0030
Epoch 31/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0029 -
val_loss: 0.0029
Epoch 32/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0029 -
```

```
val_loss: 0.0029
Epoch 33/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0028 -
val_loss: 0.0029
Epoch 34/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0028 -
val_loss: 0.0029
Epoch 35/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0028 -
val_loss: 0.0029
Epoch 36/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0028 -
val_loss: 0.0029
Epoch 37/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0028 -
val_loss: 0.0028
Epoch 38/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0028 -
val_loss: 0.0028
Epoch 39/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 40/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 41/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 42/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 43/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 44/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 45/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0027 -
val_loss: 0.0028
Epoch 46/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 47/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 48/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 49/100
```

```
2/2 [==============================] - 0s 22ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 50/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 51/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 52/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 53/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0026 -
val_loss: 0.0027
Epoch 54/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0025 -
val_loss: 0.0027
Epoch 55/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0025 -
val_loss: 0.0027
Epoch 56/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 57/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 58/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 59/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 60/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 61/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 62/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0025 -
val_loss: 0.0026
Epoch 63/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0024 -
val_loss: 0.0026
Epoch 64/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0024 -
val_loss: 0.0026
Epoch 65/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0024 -
```

```
val_loss: 0.0026
Epoch 66/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0024 -
val_loss: 0.0026
Epoch 67/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0024 -
val_loss: 0.0026
Epoch 68/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0024 -
val_loss: 0.0026
Epoch 69/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0024 -
val_loss: 0.0025
Epoch 70/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0024 -
val_loss: 0.0025
Epoch 71/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0024 -
val_loss: 0.0025
Epoch 72/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0024 -
val_loss: 0.0025
Epoch 73/100
2/2 [==============================] - 0s 25ms/step - loss: 0.0024 -
val_loss: 0.0025
Epoch 74/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 75/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 76/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 77/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 78/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 79/100
2/2 [==============================] - 0s 27ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 80/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 81/100
2/2 [==============================] - 0s 26ms/step - loss: 0.0023 -
val_loss: 0.0025
```

```
Epoch 82/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 83/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 84/100
2/2 [==============================] - 0s 25ms/step - loss: 0.0023 -
val_loss: 0.0025
Epoch 85/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0023 -
val_loss: 0.0024
Epoch 86/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0023 -
val_loss: 0.0024
Epoch 87/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 88/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 89/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 90/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 91/100
2/2 [==============================] - 0s 27ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 92/100
2/2 [==============================] - 0s 20ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 93/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 94/100
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 95/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 96/100
2/2 [==============================] - 0s 22ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 97/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 98/100
```

```
2/2 [==============================] - 0s 21ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 99/100
2/2 [==============================] - 0s 23ms/step - loss: 0.0022 -
val_loss: 0.0024
Epoch 100/100
2/2 [==============================] - 0s 24ms/step - loss: 0.0022 -
val_loss: 0.0024

Results:
              precision    recall  f1-score   support

           0       1.00      0.98      0.99      1670
           1       0.20      0.80      0.31        10

    accuracy                           0.98      1680
   macro avg       0.60      0.89      0.65      1680
weighted avg       0.99      0.98      0.99      1680

Precision: 0.1951
Recall:    0.8000
F1 Score:  0.3137
ROC-AUC:   0.9675
runtime: 0.2341 minutes
```